

HEALTHCARE • PUBLIC HEALTH ANALYSIS • POWER BI

GLOBAL HIV TRENDS & TREATMENT EFFECTIVENESS

A data-driven view of the global HIV landscape, 2000–2024

Data source: World Health Organization (WHO) | 150 countries • 6 WHO regions

The Big Picture (2024)

HIV has shifted from a fatal disease to a managed chronic condition — but the job is unfinished.

984K

NEW INFECTIONS

–59% since 2000

523K

HIV DEATHS

–70% from 2004 peak

35.2M

PEOPLE LIVING w/ HIV

Rising prevalence

79%

ART COVERAGE

PLHIV-weighted

1

Infections down, but stalling

Down 59% since 2000 — yet essentially flat from 2023 to 2024 (+0.25%).

2

Treatment is working

ART coverage and death rate move in opposite directions ($r \approx -0.48$).

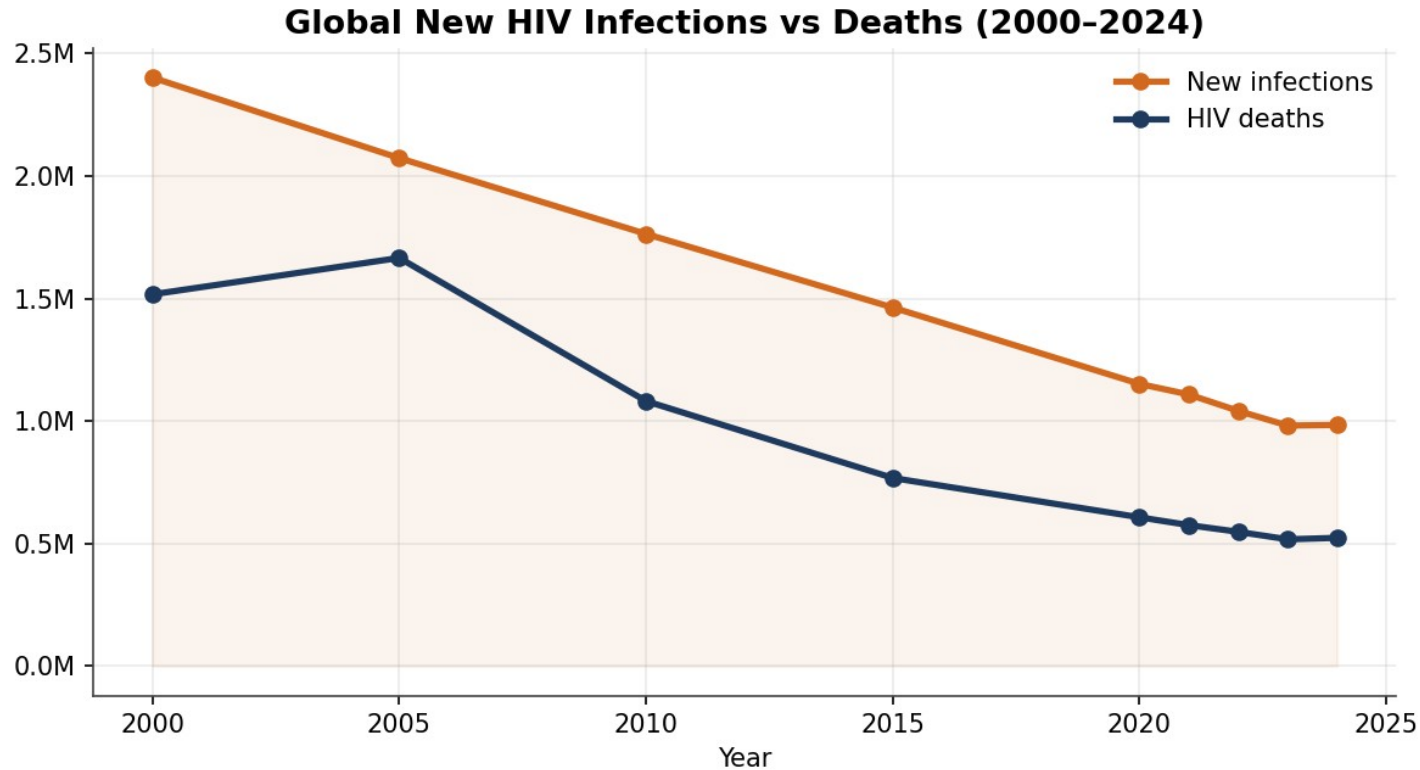
3

Burden keeps growing

More people live with HIV every year, so the care burden expands.

Are infections truly declining?

Both new infections and deaths have fallen sharply — but the decline is flattening.



What the data says

New infections: 2.40M → 0.98M

a 59% drop since 2000.

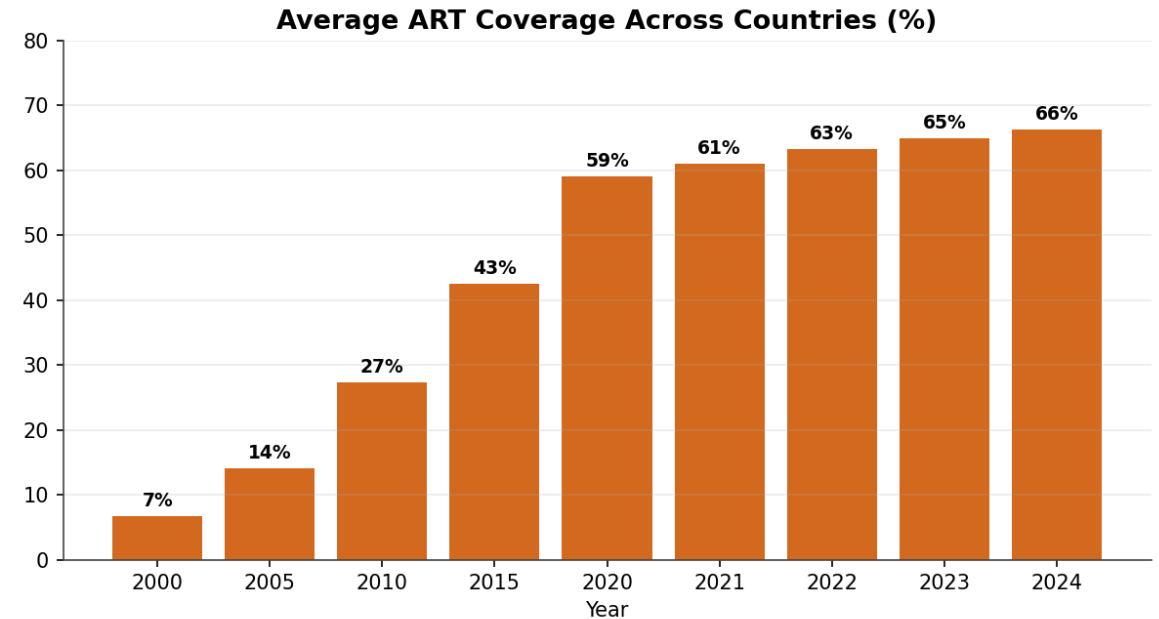
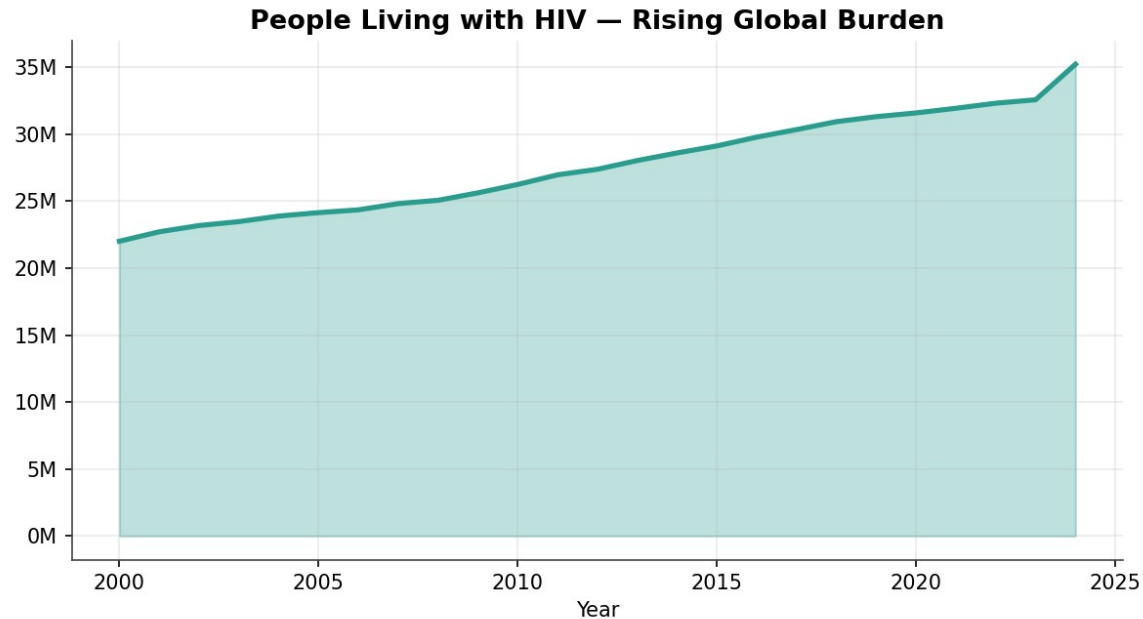
Deaths: down ~70% from the 2004 peak.

BUT: 2023→2024 infections barely moved (+0.25%).

Implication: prevention gains are plateauing — renewed effort is needed to keep the curve falling.

A success that creates a new challenge

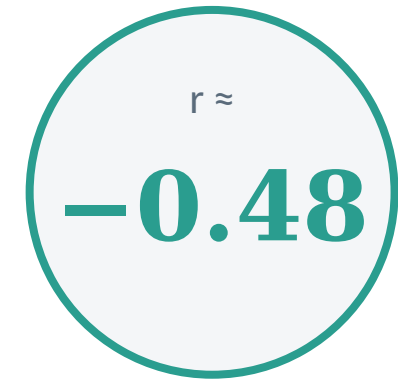
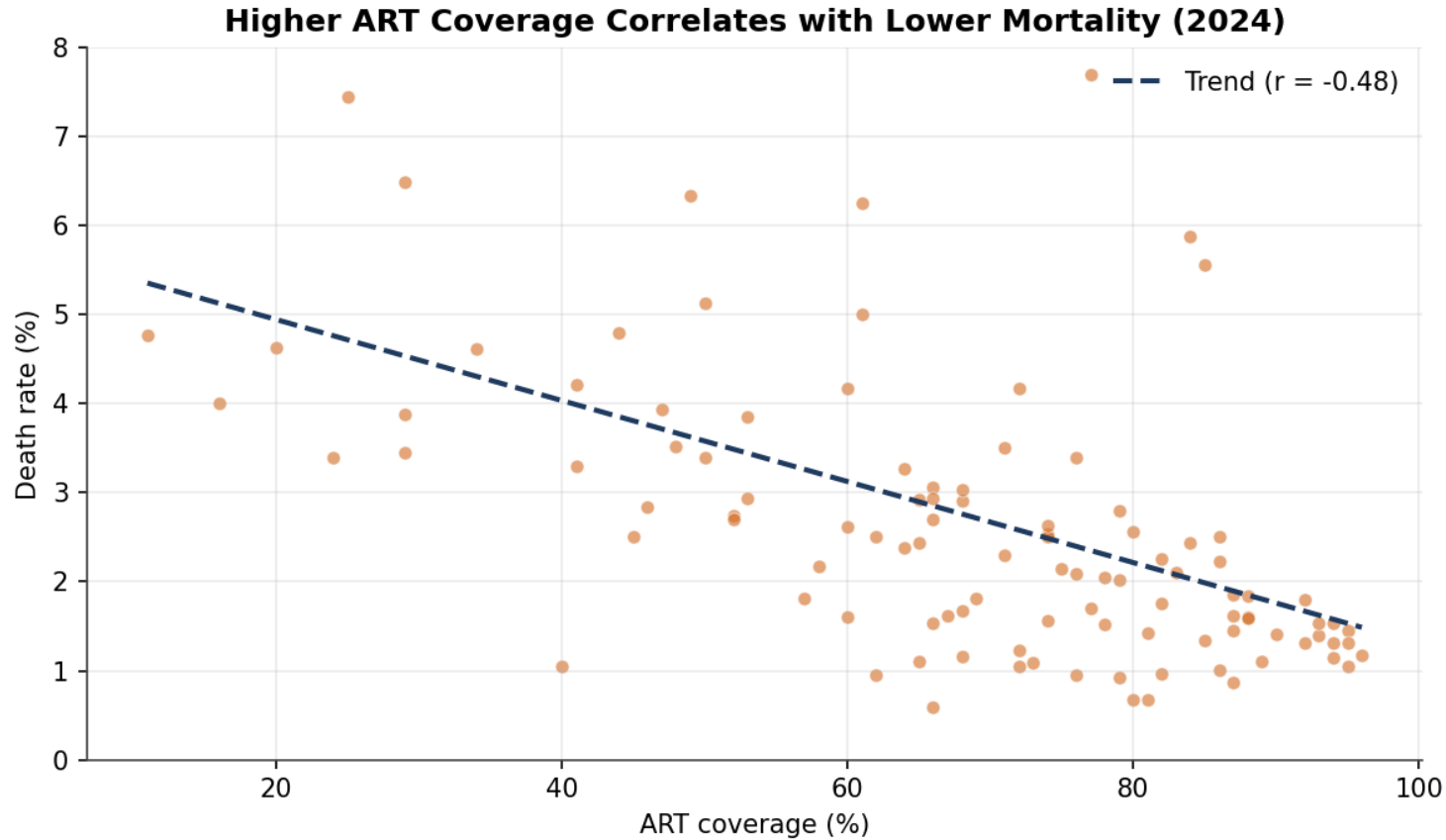
As deaths fall, more people live with HIV — driven by a ten-fold rise in treatment access.



The chronic-condition signature. People living with HIV climbed from 22M (2000) to 35.2M (2024) precisely because ART now keeps people alive — average coverage rose from 7% to 66%. The mortality win is real, but it locks in a permanent, growing need for lifelong treatment and financing.

Does treatment reduce mortality?

Across 108 countries, higher ART coverage is clearly associated with lower death rates.

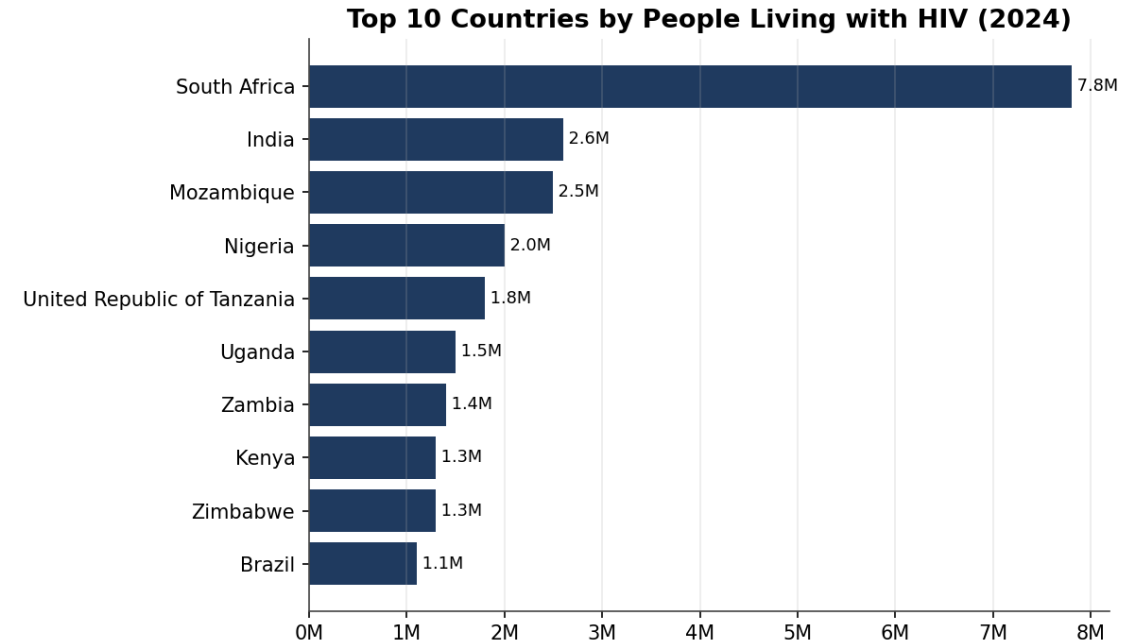
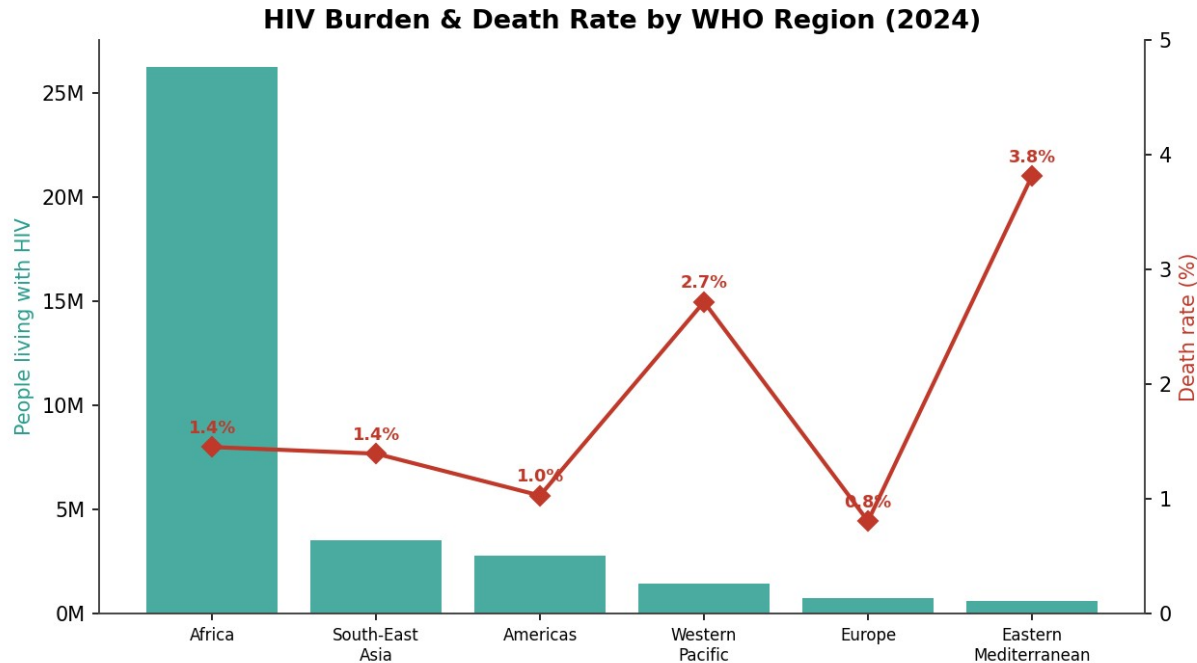


negative correlation

This is the core evidence that treatment programmes save lives. The link is moderate, not perfect, because outcomes also depend on early diagnosis, adherence, and health-system quality — but the direction is unmistakable.

The geography of the epidemic

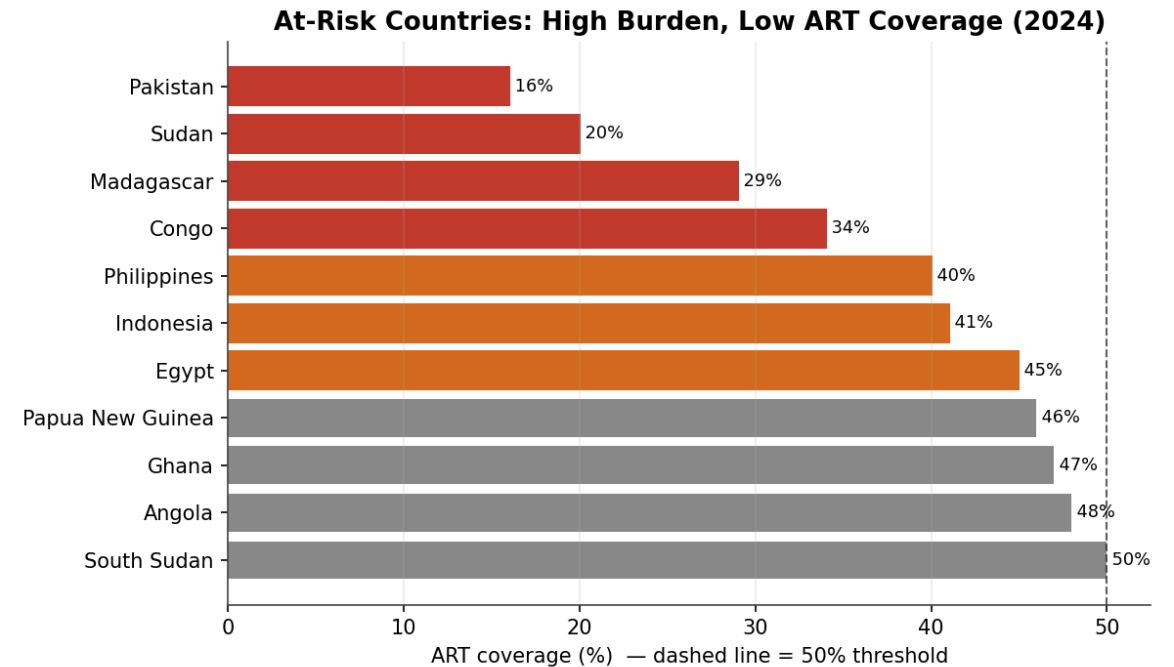
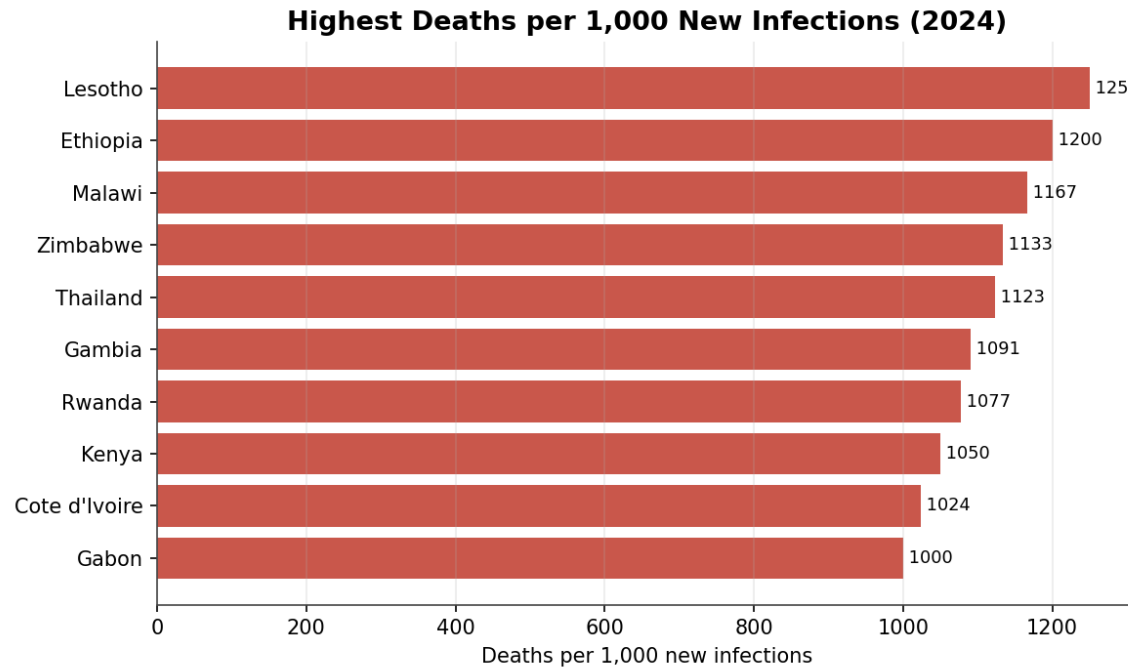
Africa carries ~75% of the global burden — but the highest death rates sit elsewhere.



Burden ≠ death rate. Africa leads on absolute burden (26.3M living with HIV, led by South Africa at 7.8M). Yet the Eastern Mediterranean (3.81%) and Western Pacific (2.71%) post the highest death rates — regions where treatment access lags the size of the epidemic.

Where intervention is most urgent

Two lenses on risk: where deaths outnumber new infections, and where coverage is dangerously low.



Priority list. Lesotho, Ethiopia, Malawi and Zimbabwe record more deaths than new infections (>1,000 per 1,000) — a retention-in-care problem. Meanwhile Pakistan (16%), Sudan (20%), Madagascar (29%) and Indonesia (41%) pair large burdens with sub-50% ART coverage — the clearest targets for urgent scale-up.

Recommendations

- 1 Close the coverage gaps**

Channel funding to low-ART, high-burden countries — Pakistan, Indonesia, Madagascar, Congo, Sudan. Largest available mortality win.
- 2 Re-energise prevention**

With infections plateauing, renew PrEP, testing and education — focused on Africa and rising-infection settings like the Philippines.
- 3 Fund the growing care pool**

Lock in sustainable, long-term financing for lifelong ART so the mortality gains are never reversed.
- 4 Target retention hotspots**

In Lesotho, Ethiopia, Malawi and Zimbabwe, prioritise early diagnosis, adherence and viral suppression.
- 5 Modernise data reporting**

Move five-yearly indicators to annual reporting everywhere for timelier monitoring and faster course-correction.

The bottom line

A public-health success in motion — with unfinished business.

Infections down 59%, deaths down 70% from peak, treatment up ten-fold. Sustained, targeted, data-driven investment — guided by the model and dashboard built here — is what keeps the curve falling.

Built with Power BI • Data: World Health Organization